SKILL-16: API Documentation for Full-Stack Student CRUD Application using Swagger

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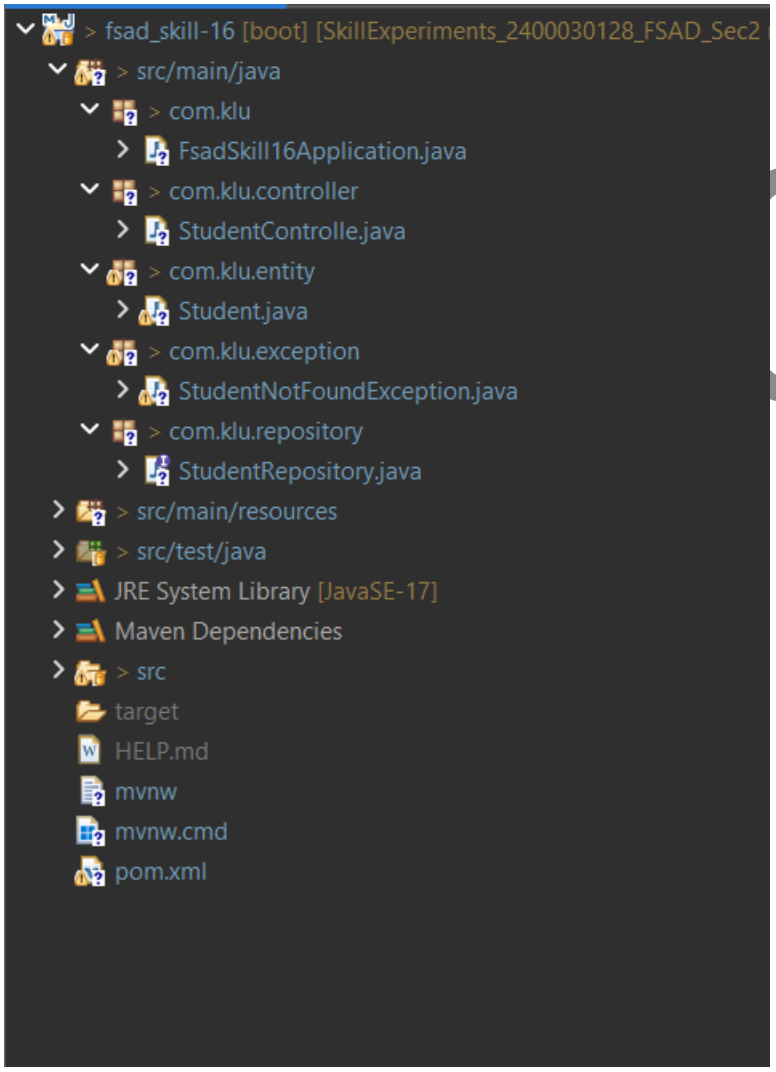
Section: 2

Scenario:

The student management system developed in SKILL – 12 includes full CRUD operations using Spring Boot REST APIs. To help testers, developers, and frontend teams understand each endpoint clearly, the backend must include Swagger/OpenAPI documentation. Swagger UI must display all Student CRUD endpoints, response structures, validation messages, and example requests.

Structure:



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Codes:

Student.java:

```
package com.klu.entity;

import org.hibernate.annotations.*;

import jakarta.persistence.*;
import jakarta.persistence.Table;

@Entity
@Table(name = "student_skill16")
public class Student
{

  @Id
  @GeneratedValue(strategy = GenerationType.IDENTITY)
  private Long id;

  private String name;
  private String email;
  private String course;

  public Long getId() {
    return id;
  }

  public void setId(Long id) {
    this.id = id;
  }

  public String getName() {
```

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```
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }

    public String getCourse() {
        return course;
    }

    public void setCourse(String course) {
        this.course = course;
    }
}
```

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StudentController.java:

```
package com.klu.controller;

import com.klu.entity.Student;
import com.klu.repository.StudentRepository;
import com.klu.exception.StudentNotFoundException;
```

```
import io.swagger.v3.oas.annotations.Operation;
import io.swagger.v3.oas.annotations.responses.ApiResponse;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
@RequestMapping("/students")
public class StudentControlle {

  @Autowired
  private StudentRepository repo;

  @Operation(summary = "Add new student")
  @ApiResponse(responseCode = "200", description = "Student created")
  @PostMapping
  public Student addStudent(@RequestBody Student student) {
    return repo.save(student);
  }

  @Operation(summary = "Get all students")
  @GetMapping
  public List<Student> getAllStudents() {
    return repo.findAll();
  }

  @Operation(summary = "Get student by ID")
  @GetMapping("/{id}")
  public Student getStudent(@PathVariable Long id) {
```

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```
    return repo.findById(id)
            .orElseThrow(() -> new StudentNotFoundException("Student not found"));
  }

  @Operation(summary = "Update student")
  @PutMapping("/{id}")
  public Student updateStudent(@PathVariable Long id, @RequestBody Student student) {

    Student existing = repo.findById(id)
            .orElseThrow(() -> new StudentNotFoundException("Student not found"));

    existing.setName(student.getName());
    existing.setEmail(student.getEmail());
    existing.setCourse(student.getCourse());

    return repo.save(existing);
  }

  @Operation(summary = "Delete student")
  @DeleteMapping("/{id}")
  public String deleteStudent(@PathVariable Long id) {
    repo.deleteById(id);
    return "Student deleted successfully";
  }
}
```

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StudentNotFoundException.java:

```
package com.klu.exception;

public class StudentNotFoundException extends RuntimeException {
  public StudentNotFoundException(String message) {
    super(message);
```

```
    }
}
```

StudentRepository.java:

```
package com.klu.repository;

import com.klu.entity.Student;
import org.springframework.data.jpa.repository.JpaRepository;

public interface StudentRepository extends JpaRepository<Student, Long> {
}
```

OUTPUTS:

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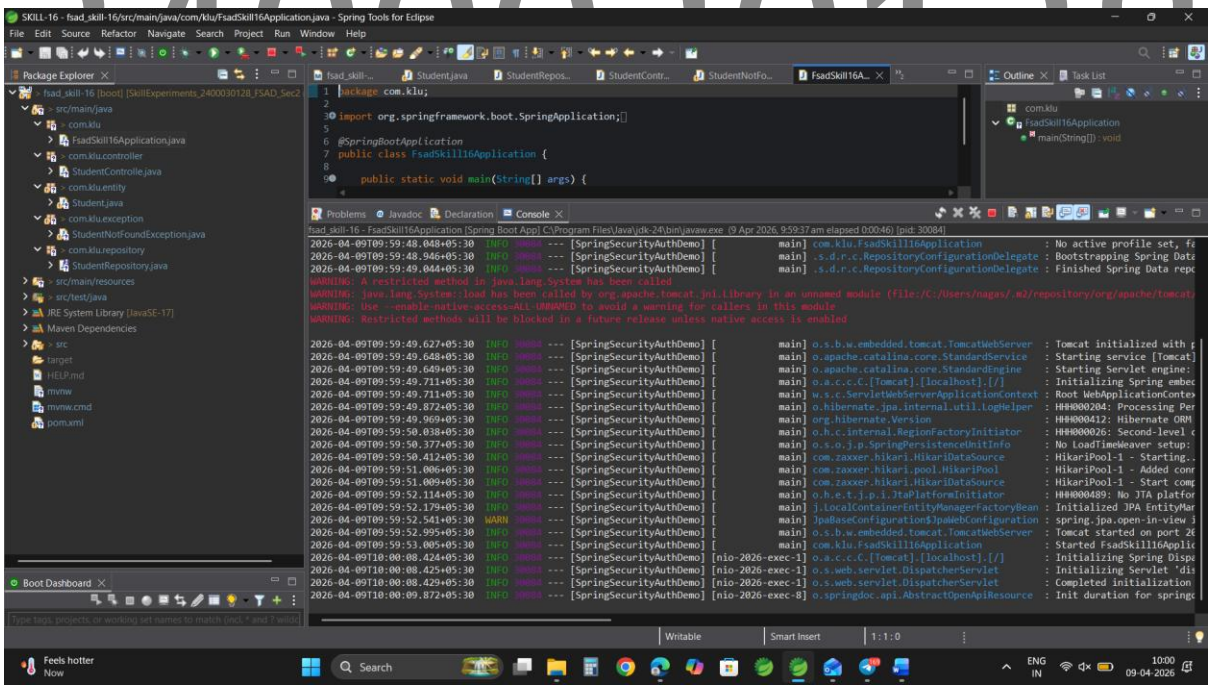
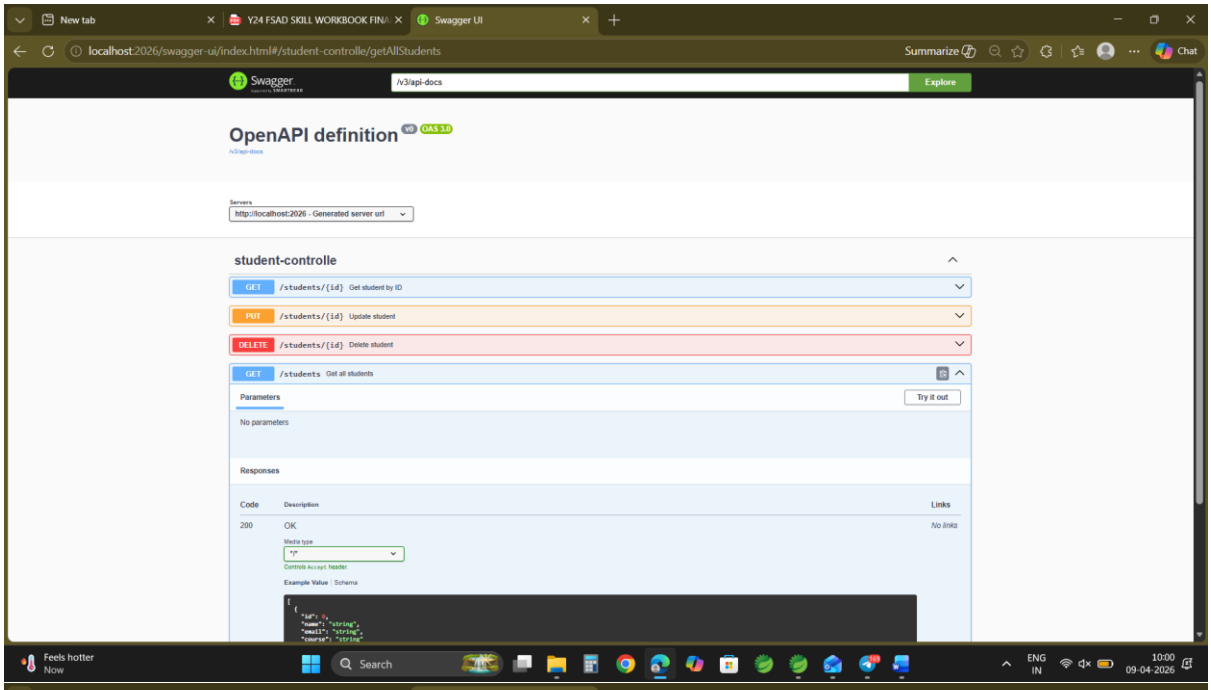
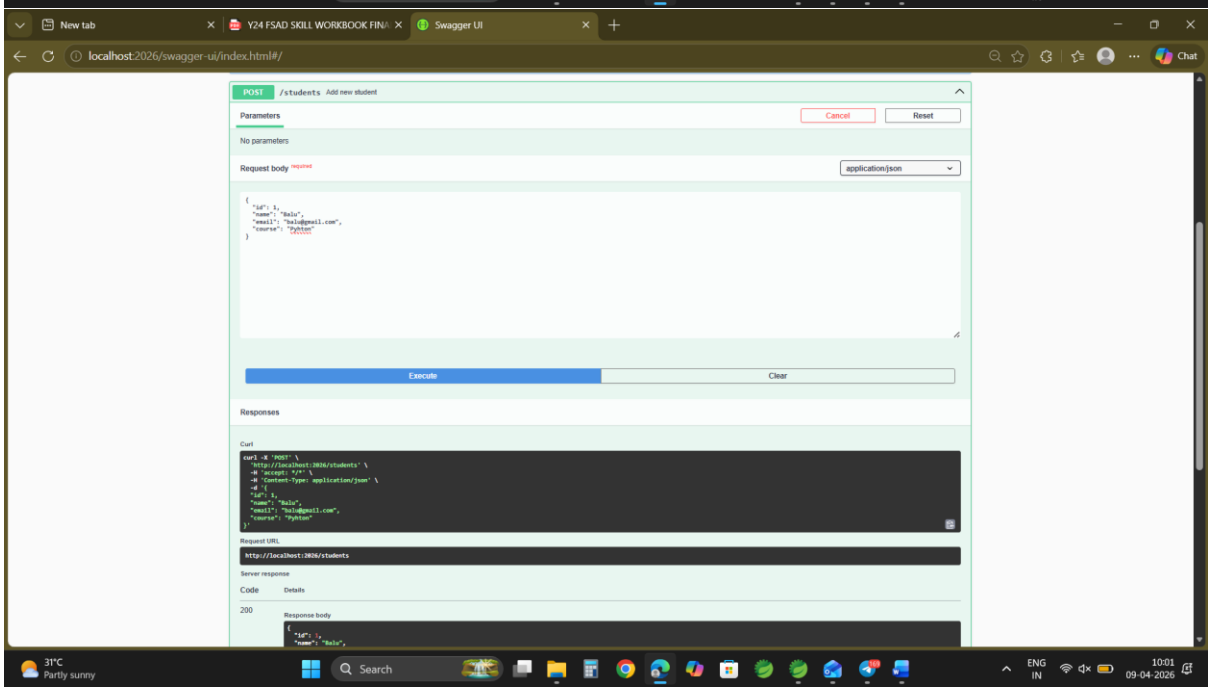



New tab
Y24 FSAD SKILL WORKBOOK FINA
Swagger UI
localhost:2026/swagger-ui/index.html#/student-controlle/getAllStudents
Summarize
Chat
Swagger
/v3/api-docs
Explore
OpenAPI definition
v0
OAS 3.0
/v3/api-docs
Servers
http://localhost:2026 - Generated server url
student-controlle
GET /students/{id} Get student by ID
PUT /students/{id} Update student
DELETE /students/{id} Delete student
GET /students Get all students
Parameters
Try it out
No parameters
Responses
Code
Description
Links
200
OK
No links
Media type
/
Controls Accept header.
Example Value | Schema
{
{
"id": 0,
"name": "string",
"email": "string",
"course": "string"
Feels hotter
Now
Search
ENG
IN
10:00
09-04-2026


New tab
Y24 FSAD SKILL WORKBOOK FINA
Swagger UI
localhost:2026/swagger-ui/index.html#/
Chat
POST /students Add new student
Parameters
Cancel
Reset
No parameters
Request body
required
application/json
{
"id": 1,
"name": "Balu",
"email": "balu@gmail.com",
"course": "Python"
}
Execute
Clear
Responses
Curl
curl -X 'POST' \
'http://localhost:2026/students' \
-H 'accept: */*' \
-H 'Content-Type: application/json' \
-d '{
"id": 1,
"name": "Balu",
"email": "balu@gmail.com",
"course": "Python"
}'
Request URL
http://localhost:2026/students
Server response
Code
Details
200
Response body
{
"id": 1,
"name": "Balu",
31°C
Partly sunny
Search
ENG
IN
10:01
09-04-2026

New tab | Y24 FSAD SKILL WORKBOOK FINA | Swagger UI

localhost:2026/swagger-ui/index.html#/

DELETE /students/{id} Delete student

GET /students Get all students

POST /students Add new student

Parameters | Cancel | Reset

No parameters

Request body (required) — application/json

```
{
  "id": 2,
  "name": "Ravi",
  "email": "ravi@gmail.com",
  "course": "Java"
}
```

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:2026/students' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "id": 2,
  "name": "Ravi",
  "email": "ravi@gmail.com",
  "course": "Java"
}'
```

Request URL

http://localhost:2026/students

Server response

Code | Details

31°C Partly sunny | Search | ENG IN | 10:01 09-04-2026

New tab | Y24 FSAD SKILL WORKBOOK FINA | Swagger UI

localhost:2026/swagger-ui/index.html#/

DELETE /students/{id} Delete student

GET /students Get all students

Parameters | Cancel

No parameters

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:2026/students' \
  -H 'accept: */*'
```

Request URL

http://localhost:2026/students

Server response

Code | Details

200

Response body

```
[
  {
    "id": 1,
    "name": "Balu",
    "email": "balu@gmail.com",
    "course": "Python"
  },
  {
    "id": 2,
    "name": "Ravi",
    "email": "ravi@gmail.com",
    "course": "Java"
  }
]
```

Download

Response headers

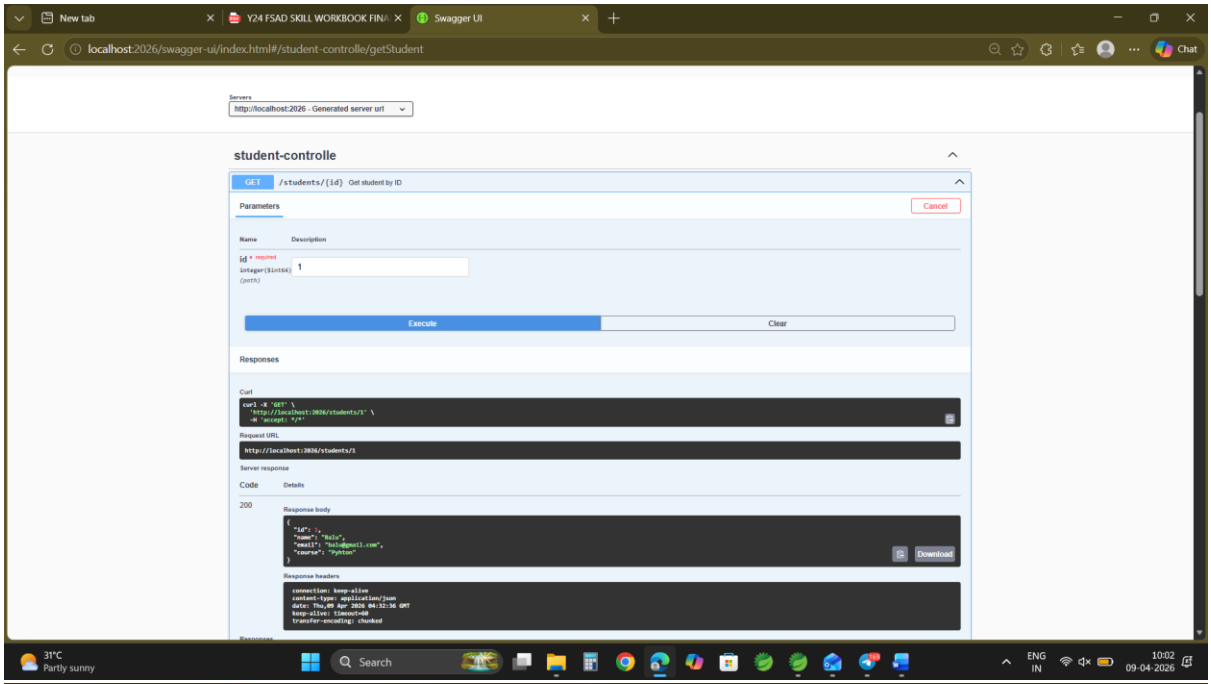
```
connection: keep-alive
content-type: application/json
date: Thu,09 Apr 2026 04:32:17 GMT
keep-alive: timeout=60
transfer-encoding: chunked
```

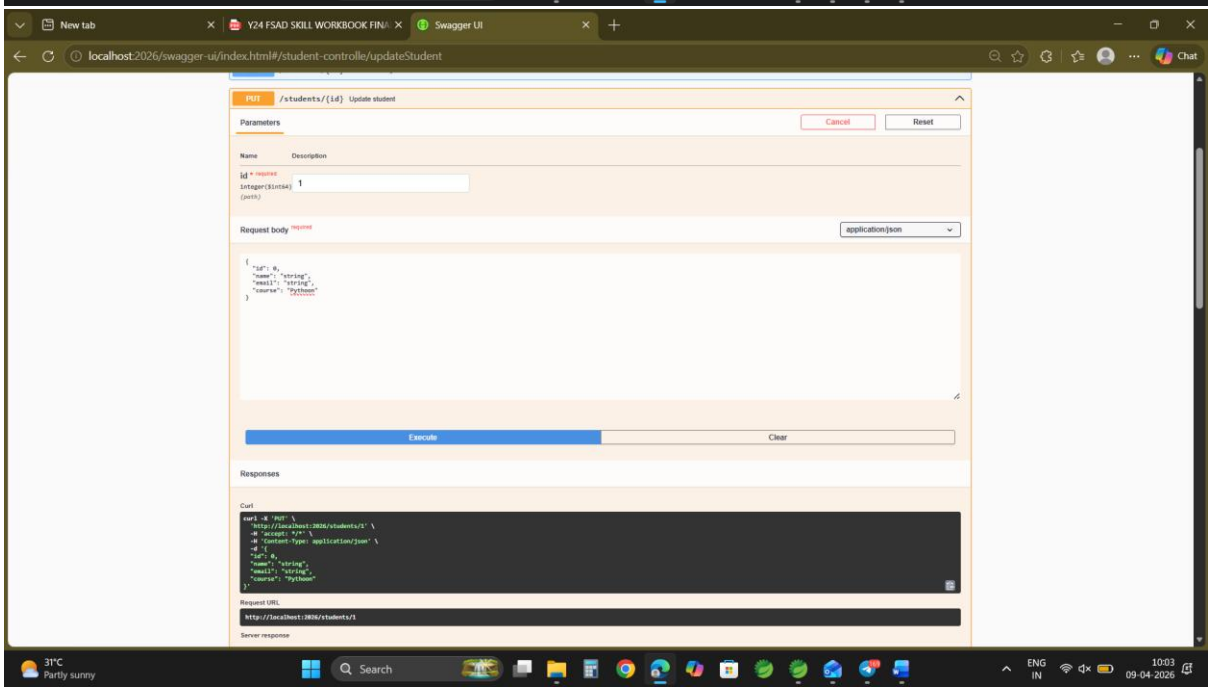
Responses

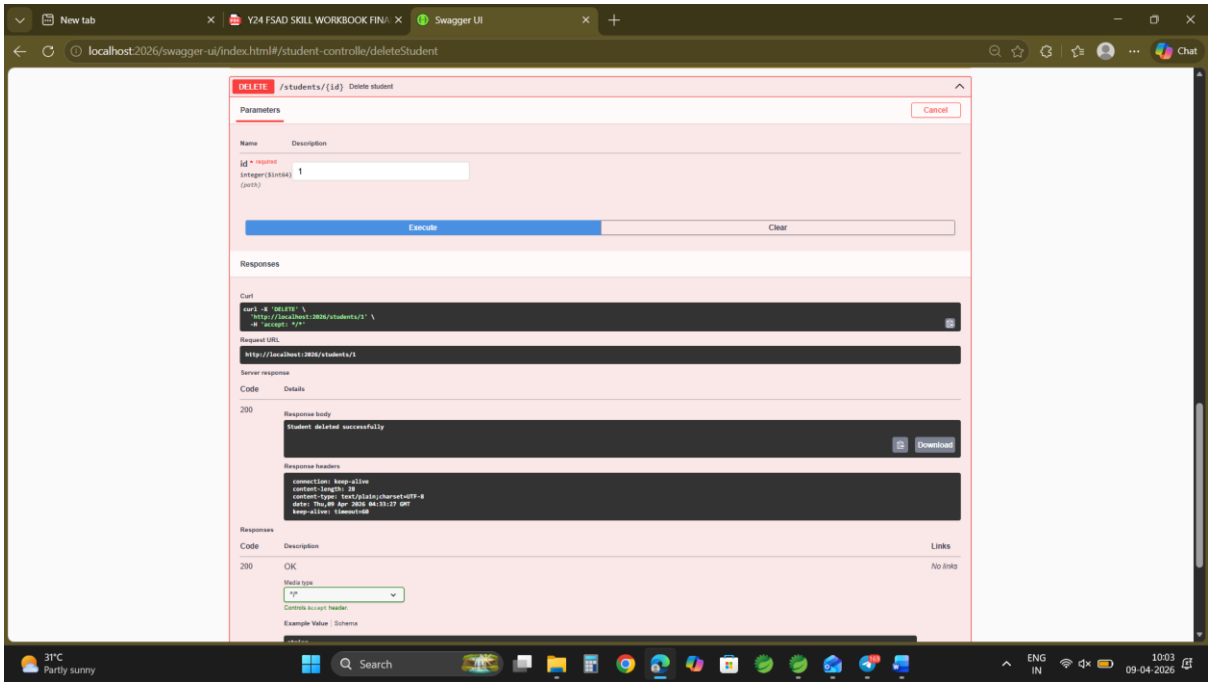
Code	Description	Links
200	OK	No links

Media type: */*

31°C Partly sunny | Search | ENG IN | 10:02 09-04-2026


New tab
Y24 FSAD SKILL WORKBOOK FIN
Swagger UI
localhost:2026/swagger-ui/index.html#/student-controlle/getStudent
Servers
http://localhost:2026 - Generated server url
student-controlle
GET /students/{id} Get student by ID
Parameters
Cancel
Name Description
id * required
integer($int64)
(path)
1
Execute
Clear
Responses
Curl
curl -X 'GET' \
'http://localhost:2026/students/1' \
-H 'accept: */*'
Request URL
http://localhost:2026/students/1
Server response
Code Details
200
Response body
{
"id": 1,
"name": "Bala",
"email": "bala@gmail.com",
"course": "Python"
}
Download
Response headers
connection: keep-alive
content-type: application/json
date: Thu,09 Apr 2026 04:31:36 GMT
keep-alive: timeout=60
transfer-encoding: chunked
31°C
Partly sunny
Search
ENG
IN
10:02
09-04-2026


New tab
Y24 FSAD SKILL WORKBOOK FIN
Swagger UI
localhost:2026/swagger-ui/index.html#/student-controlle/updateStudent
PUT /students/{id} Update student
Parameters
Cancel
Reset
Name Description
id * required
integer($int64)
(path)
1
Request body required
application/json
{
"id": 0,
"name": "string",
"email": "string",
"course": "Python"
}
Execute
Clear
Responses
Curl
curl -X 'PUT' \
'http://localhost:2026/students/1' \
-H 'accept: */*' \
-H 'Content-Type: application/json' \
-d '{
"id": 0,
"name": "string",
"email": "string",
"course": "Python"
}'
Request URL
http://localhost:2026/students/1
Server response
31°C
Partly sunny
Search
ENG
IN
10:03
09-04-2026



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GITHUB LINK:

https://github.com/nagasubrahmanyammeka/SkillExperiments_2400030128_FSAD_Sec2/tree/main/SKILL-16